

Binary Distillation

Final Lab Report
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Group 1/Tuesday/PM

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Executive Summary

The objective of this experiment was to determine the variation with boil-up rate of pressure drop over the distillation column and to determine overall column efficiency. Additionally, to carry out the distillation at constant reflux ratios by varying top and bottom compositions by time. In experiment A, the boil-up rate was calculated and determined that as boil-up rate increases as a function of pressure drop, the change of the power levels 0.5, 0.75, 1 and 1.25 kW of the system also increases. The efficiency of the column is calculated from the theoretical plates and the actual plates. The average value of the efficiency of the column was 84.57% which is very high in comparison to the full-scale columns used in the industry where efficiency is 50 – 70 %. The reason for such unreasonable efficiency in this experiment is due to human experimental error. In experiment B, reflux ratios of 5/1 and 10/1 ratio are observed, both with the power set at 0.75 kW. Parameters are found and calculated to construct McCabe-Thiele diagrams for both reflux ratios. In the 5/1 reflux ratio, the liquid molar fraction of ethanol in the distillate is 0.8710. The VLE literature data, the rectifying line, and the 45-degree line all “pinch” indicating an infinite number of trays required in order to obtain this amount of separation. In the 10/1 reflux ratio, the molar fraction of ethanol in the distillate is noted to be 0.8784, again indicating the need for an infinite number of trays. The results obtained are not what is expected indicating a high level of error in the experimental process. The high tower efficiency calculated with the data in this experiment does not coincide with literature values and reality. Moreover, the high recovery of ethanol in the distillate is also unreasonable according to the literature.

Introduction

The overall objective of this experiment is to use the Armfield UOP3CC Continuous Distillation Column to apply the McCabe-Thiele method to the separation of ethanol and water. This requires us to determine the variation with boil-up rate of pressure drop over the distillation column and to determine overall column efficiency. Additionally, it is necessary to carry out the distillation at constant reflux ratios by varying top and bottom compositions by time. Distillation is a very useful method to efficiently separate the more volatile component from a liquid mixture of the light and heavy components in the feed. During this process, the lighter component is vaporized and leaves the column through the distillate whereas the heavier component is left as a liquid in the bottoms stream of the column. This technique is widely chosen for its high degree of separation. [1] The significance of this experiment allows us to understand how this binary distillation works to separate two components from each other by analyzing and determining the boil-up rates, column efficiency, column pressure drop, and the product composition at different boiler power and reflux ratios.

For part A of the experiment, an ethanol and water mixture consisting of 1.2L of ethanol and 8.8L of water is made and filled into the reboiler and heated above the boiling point to undergo phase changes within the column. This experiment will operate at a total reflux by switching off the reflux ratio timer. On the console, the temperature selector switch was set at T1 and T8 along with selecting a reboiler heating element to 1.5kW. When total reflux occurred, the power was then set

to 0.5kW and parameters such as pressure drop from the pressure drop meter, refractive index from the refractometer, and temperatures at T1 and T8 were measured when equilibrium was reached. The boil up rate was varied up to 1.25kW and same measurements were taken in between. For part B in the experiment, the heat controller was set at 0.75kW and after equilibrium the reflux controller was used to set two different reflux ratios. The condenser condenses the vapor from the distillation column. The decanter collects the condensed liquid and sends it to the reflux valve that will allow the condensate to flow to column or receiver. Timed volume collection was used to get samples from valve 3 and valve 2 in the apparatus. Data for parameters such as refractive index to determine component compositions and temperatures at T1 and T8 were also obtained.

This experiment showcases important parts of chemical processes that involve separation of binary mixtures which can be key in obtaining single chemical components. Distillation processes have very efficient separation techniques that separate components based on their boiling points. Additionally, distillation methods are very important technologies in the production of gasoline, diesel, and other fuels and chemicals all from naphtha feedstocks. [2] Distillation is also used in obtaining clean water and stronger alcoholic drinks. Distillation helps remove impurities from unclean water and increases concentration of alcohol by separating the excess water from the alcohol in the mixture. [2] Distillation is also a huge factor in medicine. The purity and quality of chemical components are super important, and distillation helps control those parameters by separating heavy impurities from the desired drug. [2] Most industrial applications are in the petroleum oil and gas, petrochemical, and natural gas industry.

Theory

In a distillation column, a mixture is boiled through heat supplied in the reboiler and the corresponding vapor flows up the column and the liquid flows down the column. At the top of the column, the product vapors are condensed, and reflux distillate is returned to the column. For a process with no reflux, the condensed liquid is not returned to the column. A portion of the bottom liquid is also returned but as vapor due to the reboiler reheating the liquid. [4] The reflux ratio is essential to controlling the product purity and determining the energy required by the reboiler. [3]

Inside the column are stages where gas and liquid make contact on a plate. There are n number of plates and ideally, the vapor, V , and liquid, L , leaving a plate are in equilibrium, whereas the streams entering are not. The streams entering move towards an equilibrium state making the streams leaving theoretically in equilibrium. This concept is demonstrated in Figure 1 in which two streams leave the plate and two streams enter the plate. Equilibrium concentrations of the liquid and vapor are denoted by x and y . The more volatile component is vaporized first and moves up the column. Any liquid moving down the column is continuously vaporized to remove the more volatile component and decreases the liquid concentration. Conversely, the vapor moving up the column is continuously condensed and increases the y concentration. [4] At total reflux, the liquid flow is equal to the vapor flow inside the column, thereby making the material balance over the most volatile component:

$$Vy_n = Lx_{n+1} \quad (1)$$

$$y_n = x_{n+1} \quad (2)$$

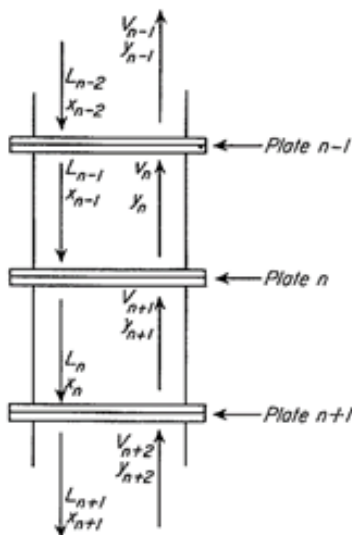


Figure 1. Material balance diagram of column. [4]

Although equilibrium is assumed, it is not achieved in practice. To measure the closeness of the concentrations to equilibrium, the overall column efficiency is determined using the following equation:

$$E = \frac{\text{Number of theoretical plates}}{\text{Number of actual plates}} \times 100 \quad (3)$$

Calculating the efficiency requires the composition of the distillate, d, and bottoms, b, liquid products and the volatilities, α , of each component. Samples of distillate and bottoms products are taken over a range of boil-up rates, in liters per hour, and the compositions are measured using a refractometer. Once the liquid mol fractions, x, and average relative volatility are known, the theoretical number of plates, n, at total reflux are calculated using the Fenske equation:

$$n + 1 = \frac{\log \left[\left(\frac{x_A}{x_B} \right)_d \left(\frac{x_B}{x_A} \right)_b \right]}{\log(\alpha_{A/B})_{avg}} \quad (4)$$

$$\text{Average relative volatility: } (\alpha_{A/B})_{avg} = \sqrt{\alpha_d \alpha_b} \quad (5)$$

A is the more volatile component and B is the least volatile component. For a distillation column containing a certain number of plates plus the reboiler, certain compositions are calculated that are shown to be incorrect through experimentation. [3]

McCabe-Thiele analysis is a graphical method for designing a distillation column for a binary system. Vapor-liquid equilibrium (VLE) data is plotted as a y vs. x diagram to create a curved

equilibrium line. Also, a linear line of x equals y is plotted. The operating line is plotted as a single line, for batch distillation, or as two separate lines for continuous distillation with a feed. Molar flows of gas and liquid are constant throughout the column in batch distillation, resulting in a single line. Columns with a feed result in one line that represents the rectifying section, which is above the feed, and the stripping section, which is below the feed. [3]

To determine the number of stages, a step structure is drawn between the operating and equilibrium lines, starting at x_D , or the concentration of distillate product, on the operating line and moving down to x_B , or the concentration of the bottom product. Figure 2 exhibits the step structure in which each step represents a stage. [4]

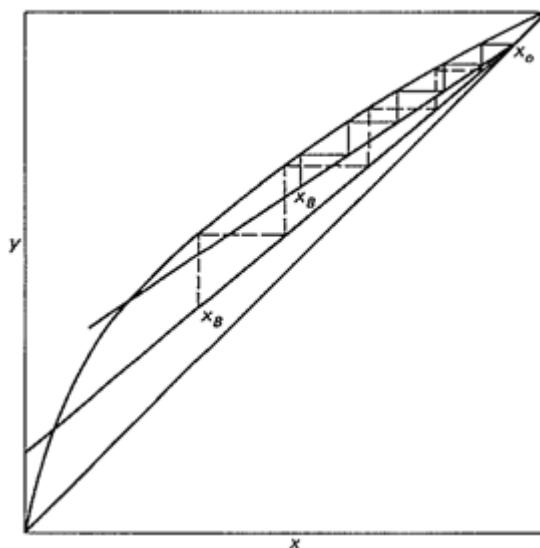


Figure 2. McCabe-Thiele diagram for batch distillation. [4]

One factor that effects the number of stages is the reflux ratio which can change the rectification operating line. An increase in the reflux results in the operating line moving closer to 1, the max value of composition. When this occurs, more of the liquid rich in the more volatile component is being recycled back into the column. This means better separation and less trays are required to obtain that separation. However, a decrease in reflux moves the line towards the equilibrium line, increasing the number of trays required. [5] The reflux ratio is essential to controlling the product purity and determining the energy required by the reboiler. [4]

Another factor effecting the design of distillation columns is pressure drop. [5] Sieve trays, or plates, contain holes that restrict the vapor and liquid flowing through the column. Total pressure drop is the sum of the pressure caused by this restriction. There is an overall increase in pressure drop when the velocity of the vapors moving up the column increases. Different pressure drops are measured by varying the velocity, which is controlled by varying the boil-up rate, which is in turn controlled by the power to the reboiler. When the head of the liquid is overcome and foaming occurs, the pressure drop is proportional to the square of the velocity. Graphing a log of pressure drop and boil-up rate shows a constant pressure drop at low boil-up rates. The occurrence of

foaming from increased boil-up rate shows a sharp rise in pressure. [3] This relationship is displayed in Figure 3. [6]

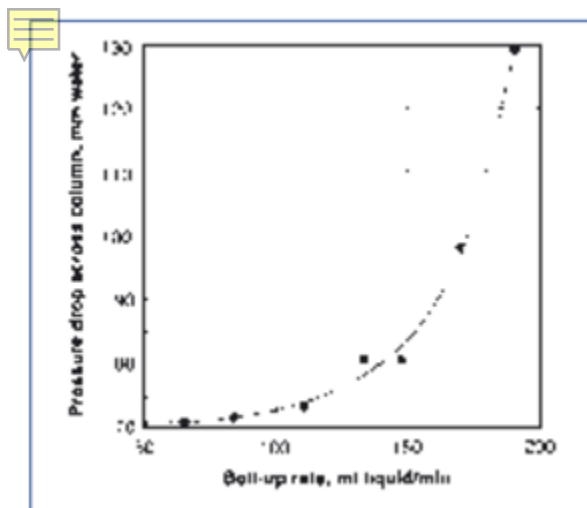


Figure 3. Column pressure drop as a function of boil-up rate. [6]

Results

Calibration Curve

Ten solutions of increasing molar composition of ethanol in water are prepared, and respective refractive indices are measured to construct a calibration curve to calculate the percent molar concentration of each sample from the column. Each concentration is graphed versus the corresponding index in Excel and a best fit second order polynomial curve is fit to the data with the corresponding equation displayed. A best fit curve for weight percent ethanol vs refractive index is made as shown in Figure 4. This is converted to a molar fraction of ethanol versus refractive index using the densities and molecular weights of water and ethanol. This curve is shown in Figure 5. See Table 4 in Appendix I for raw data.

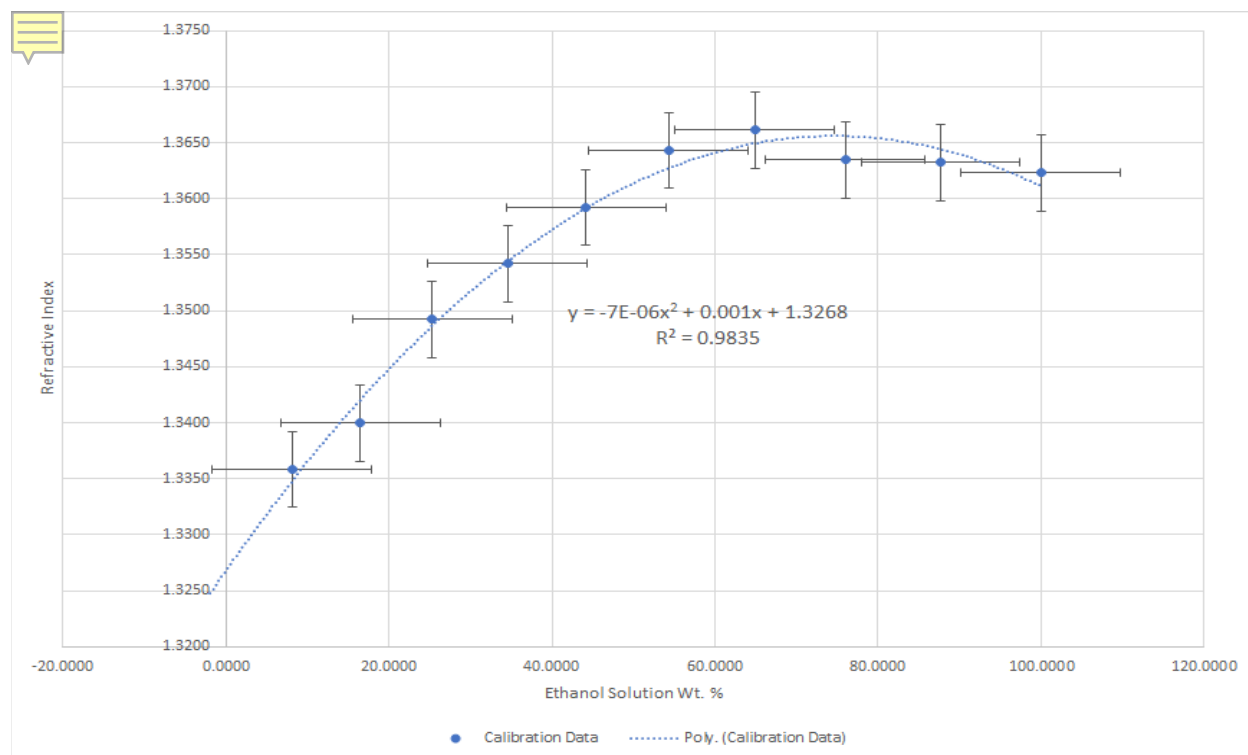


Figure 4. Calibration curve for weight percent of ethanol vs refractive index with corresponding best fit second order polynomial equation

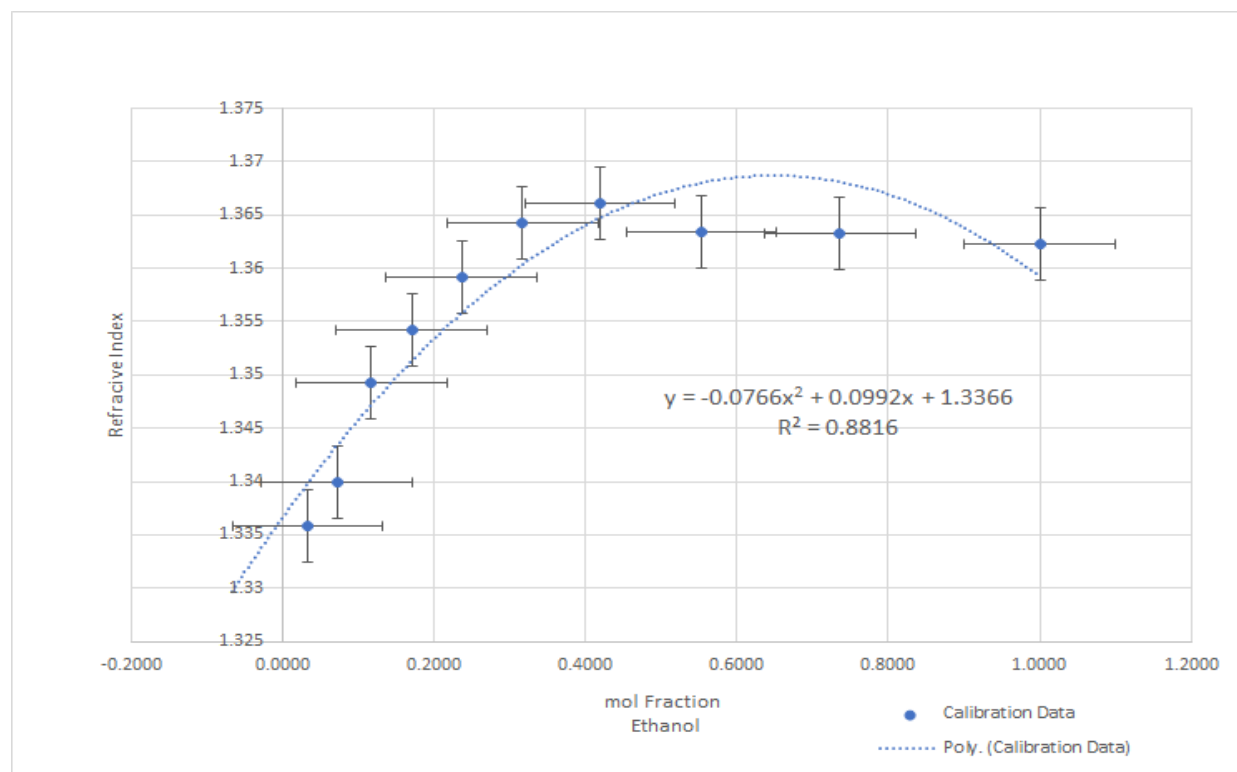


Figure 5. Calibration curve for molar fraction of ethanol vs refractive index with corresponding best fit second order polynomial equation

The regression analysis R-squared value is displayed on each graph. The weight percent calibration curve displays a R-squared value of 0.9835 which indicates very low variance and good curve fit. The R-squared value of 0.8816 on the molar concentration graph indicates higher variance.

Experiment A: Effect of Boil-Up Rate and Overall Column Efficiency

Experiment A examines changes of the boil-up rate with the pressure drop and the efficiency of the distillation column. The data collected is used to calculate boil-up rates in power levels of 0.5 kW, 0.75 kW, 1.0 kW and 1.25 kW, with the resulting liquid samples of the overhead and bottoms of the distillation column.

The boil-up rate is calculated from the volume of the sample collected in relation to its time rate. The data collected from the samples is presented in detail in tables 5 and 6, appendix I, and the following table 3 represents the computed boil-up rates:

Table 1: Boil-up rate results with different pressure drop

Power(kW)	Boil-up Rate (L/hr)	Overall dP
0.5	1.02	67
0.75	2.38	68
1	3.42	74
1.25	4.23	105

The resulting rate is plotted into the following figure:

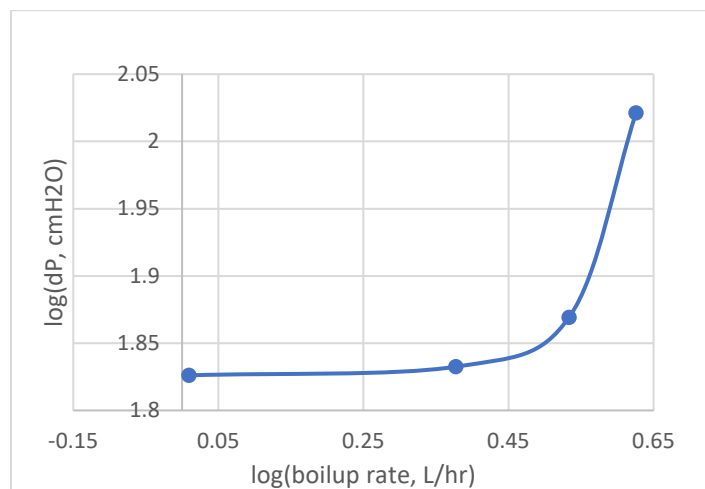


Figure 6. relation of boil-up rate to the pressure drop

The figure 6 demonstrates the dependence of the boiler rate increase as the function of the pressure drop, that is induced by the change of the power levels of the system. At first it demonstrates constant pressure drop at lower boil-up rates followed by the sudden increase of the boil-up rate with the increase of the pressure drop.

The visual observations of the bubbling in the column are recorded and presented in the following list:

Power(kW) degree of foaming (on trays)

0.5	very gentle
0.75	gentle localized
1	violently localized
1.25	violently foaming

The overall column efficiency results are obtained from the calculations of the overhead composition of the $x(\text{EtOH})_d$, and the bottoms composition $x(\text{EtOH})_b$.

The overall efficiency of the column is computed from the following equation:

$$\eta = \frac{\text{Number of theoretical plates } (N)}{\text{Number of actual plates}} * 100$$

The Fenske equation is used to find number of the theoretical plates N :

$$N = \frac{\log \left[\left(\frac{x_{\text{EtOH}}}{x_{\text{H}_2\text{O}}} \right)_d \left(\frac{x_{\text{H}_2\text{O}}}{x_{\text{EtOH}}} \right)_b \right]}{\log(\alpha_{\text{EtOH}/\text{H}_2\text{O}})_{\text{avg}}} - 1$$

To compute the relative volatility $\alpha_{\text{EtOH}/\text{H}_2\text{O}}$ from the Fenske equation above, the subsequent relation is used:

$$\alpha_{\text{EtOH}} = \frac{\frac{y_{\text{EtOH}}}{x_{\text{EtOH}}}}{(1 - y_{\text{EtOH}})/(1 - x_{\text{EtOH}})}$$

Next, the values of the vapor fractions y_{EtOH} are found from the water/ethanol equilibrium data (VLE data) from the literature [5], by interpolation of the experimental data x_{EtOH} with the VLE data. The interpolation equation used is:

$$y_{\text{EtOH}} = y_1 + (x_{\text{EtOH}} - x_1) \frac{y_2 - y_1}{x_2 - x_1}$$

The results of the efficiency of the column are listed in the following table:

Table 2: Results of the efficiency of the column

Power (kW)	Distillate		Bottoms		α			y (water)	y (ethanol)	N	Efficiency %
	x(EtOH)	x (water)	x (ethanol)	x (water)	Distillate	Bottoms	Average				
0.5	0.9117	0.0883	0.0165	0.9835	1.0000	0.0969	0.3159	0.8524	0.9117	6.57	82.17
0.75	0.8880	0.1120	0.0149	0.9851	1.0150	0.0983	0.3159	0.8667	0.8895	6.43	80.38
1	0.8880	0.1120	0.0115	0.9885	1.1759	0.1851	0.3208	0.8971	0.8895	6.74	84.24
1.25	0.8880	0.1120	0.0068	0.9932	1.1887	0.1840	0.3275	0.9392	0.8895	7.32	91.50
									Average	6.77	84.57

The efficiency is calculated with 8 actual plates, while the number of the theoretical plates are 7. The average value of the efficiency of the column is 84.57%.

The additional data analysis:

To analyze when does the column achieve steady state, the temperature of the top plate T1 and the temperature of the bottom plate T8 is observed, while their average value is used as the temperature of the column. Figure 7 shows that the temperature was increasing to 90 °C followed by the decrease to 80 °C, where the steady state of the column is achieved. The figure 11 with the detailed temperature vs elapsed time plot is listed in the appendix I.

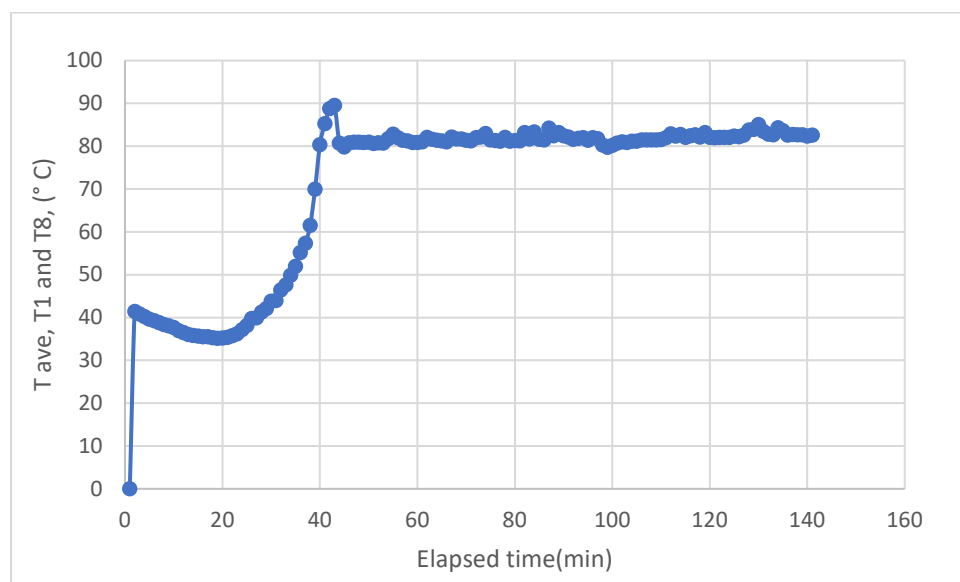


Figure 7. Average temperature of the column vs elapsed time

The column is visually observed to change from the “gentle localized” to “violent localized” in the temperature until 70°C, and “foaming gently over whole tray” up to “violently foaming over whole tray” in the area of 90°C. In the steady state temperature where the T1 and T8 achieved the same values, tray was visually “violently foaming.” There is a direct relation between the boil-up rate and the column efficiency, so that if the boil-up rate is too high or too low, the efficiency of the column will be low as well. It is often referred as the column being “too dry” or “the liquid flooding in the column” or entrainment of liquid. [9]

Experiment B: Distillation at a Constant Reflux Ratio

Two reflux ratios are observed in this section including a 5/1 ratio and a 10/1 ratio, both with the power set at 0.75 kW. For each reflux ratio, 3 samples are taken from the top of the column every 10 minutes. The volume of each sample and time it takes to collect that volume is recorded. One sample from the bottom is obtained. The refractive indices are measured, and molar composition calculated for each sample. The boil-up rate is calculated. See Table 3.

Table 3. Reflux ratio data to construct McCabe-Thiele diagram

Reflux Ratio (L/D)	X _d	X _B	Boil-up Rate (l/hr)	Process Temp (°C)	Column Temp (°C)
10 to 1	0.8784	0.0177	2.44	92.7	82.4
5 to 1	0.871	0.0122	2.06	93.1	82.4

With this data, a McCabe-Thiele diagram can be constructed. The vapor-liquid equilibrium (VLE) data for methanol and water mixtures from the literature is graphed along with the 45-degree equilibrium line, the “rectifying” line, and the “stripping” line. [7] “The operating lines define the mass balance relationships between the liquid and vapor phases in the column. There is one operating line for the bottom (stripping) section of the column and one for the top (rectifying) section of the column.” [4] Since the process examined in this lab is a batch process, only the rectifying line is needed. The symbol “x” denotes the composition of the component in the liquid phase and “y” denotes the composition of the component in the vapor phase. y_{n+1} denotes the vapor composition of the component in the tray below tray ‘n’.

The “rectifying” line is determined by using the reflux ratio and molar composition of ethanol from the distillate. The general equation for the “rectifying” line is as follows:

$$\frac{\frac{L}{D}}{\frac{L}{D}+1} * x_n + \left(1 - \frac{\frac{L}{D}}{\frac{L}{D}+1}\right) * x_d = y_{n+1}$$

Where x_d is the molar fraction of ethanol in the distillate and L/D is the reflux ratio. .

The rectifying equation for reflux ratio 5/1 is $y_{n+1} = \frac{5}{6}x_n + 0.145167$.

The rectifying equation for reflux ratio 10/1 is $y_{n+1} = \frac{10}{11}x_n + 0.079855$.

See Tables 8 and 9 in Appendix I for data to graph each diagram in Figures 8 and 9. See Appendix I for all raw data.

Figure 8 is the McCabe-Thiele graph for reflux ratio 5/1. The liquid molar fraction of ethanol in the distillate is 0.8710. At this point on the graph, the VLE literature data, the rectifying line, and the 45-degree line all “pinch” indicating an infinite number of trays required in order to obtain this amount of separation. To illustrate how the trays are “stepped” off on the diagram, trays are started with a liquid molar fraction in the distillate (x_d) at about 0.79. From here, 8 trays plus a reboiler are required, which is what the physical tower contains.

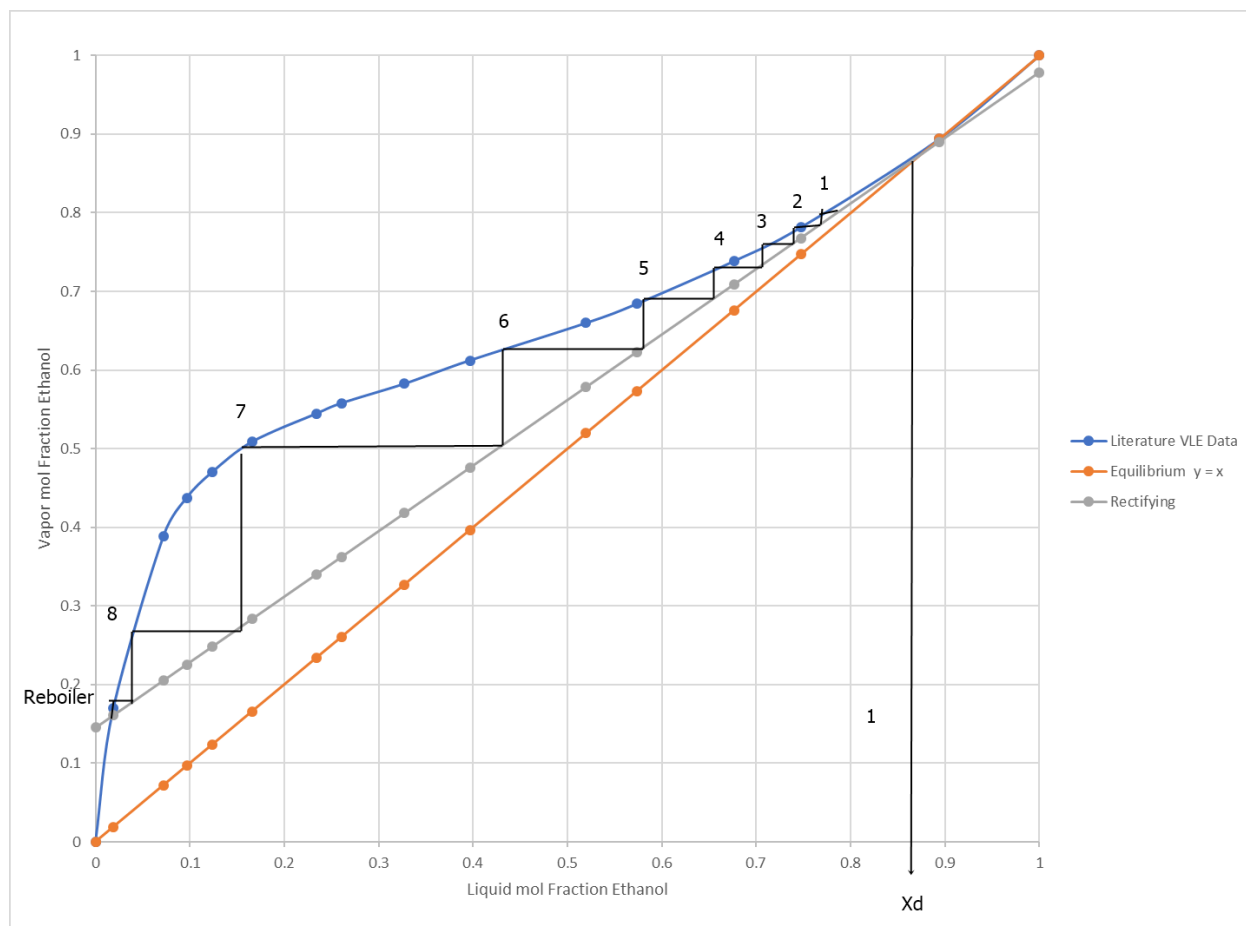


Figure 8. McCabe-Thiele diagram for reflux ratio 5:1.

Figure 9 is the McCabe-Thiele diagram for reflux ratio 10/1. The molar fraction of ethanol in the distillate is noted to be 0.8784. The graph is nearing a “pinch” at this point, again indicating an infinite number of trays required. Trays are “stepped-off” at an ethanol mol fraction of 0.81 in the distillate to obtain 8 trays plus a reboiler required.

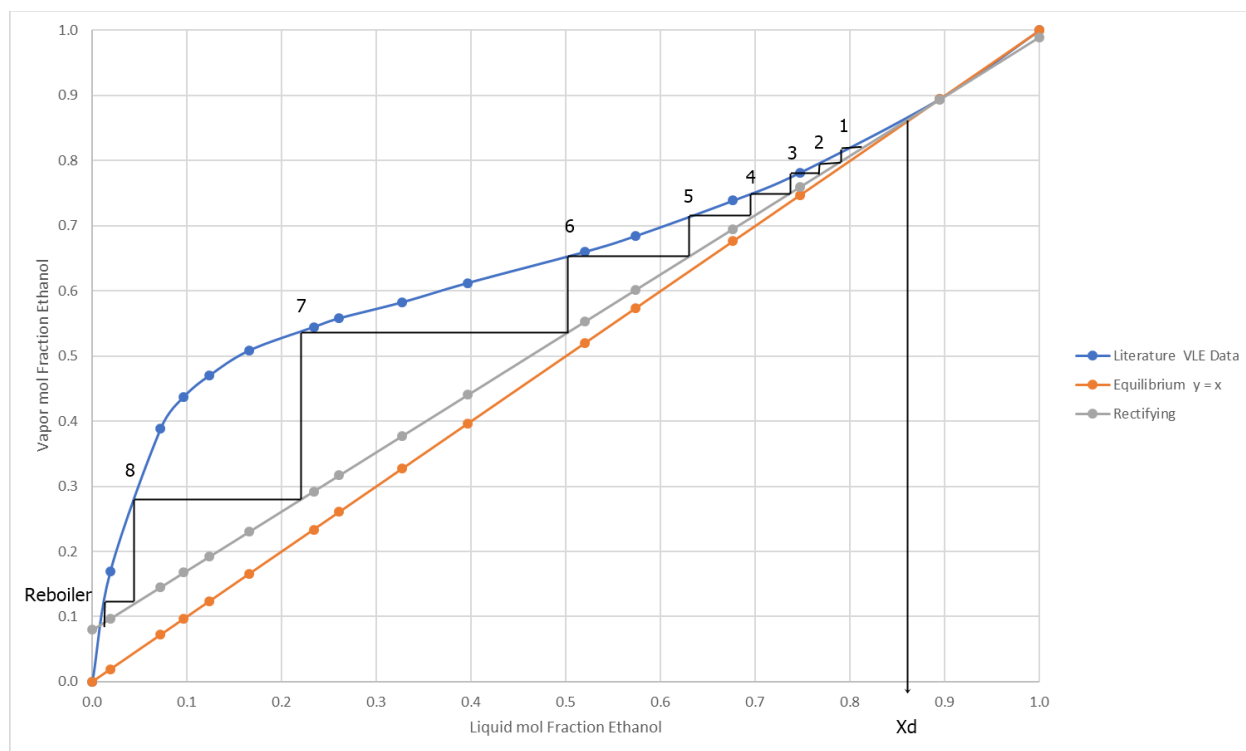


Figure 9. McCabe-Thiele diagram for reflux ratio 10:1

Discussion

This experiment is completed to determine the efficiency of a distillation tower in separating ethanol and water mixture at various reflux ratios and power settings using the McCabe-Thiele method. Analysis of the ethanol-water distillation system is mathematically straightforward when using molar quantities rather than the more common measurements of volume or weight. This is because of an energy balance principle called constant molar overflow. Essentially, this principle states that the heat (energy) required to vaporize or condense a mole of ethanol is approximately equal to the heat (energy) required to vaporize or condense a mole of water; and is approximately equal to the heat (energy) required to vaporize or condense any mixture of the two.” [8] The results obtained are not what is expected indicating a high level of error in the experimental process.

The high tower efficiency calculated with the data in this experiment does not coincide with literature values and reality. “Typical values for tray efficiency range from 0.5 to 0.7 and depends on a number of factors, such as the type of trays being used, and internal liquid and vapor flow conditions. Sometimes, additional trays are added (up to 10%) to accommodate the possibility that the column may be under-designed.” [4]. The greater than 80% column efficiency determined for the column in this lab is very high and indicates errors within the experimental process.

The high recovery of ethanol in the distillate is also unreasonable according to the literature. The VLE curve contacts the 45-degree line at around 0.9 mol fraction ethanol. This is a point that

indicates an azeotrope, and the water and ethanol cannot be separated even if an infinitely large tower was made with an infinitely large amount of energy supplied. [8] See Figure 10.

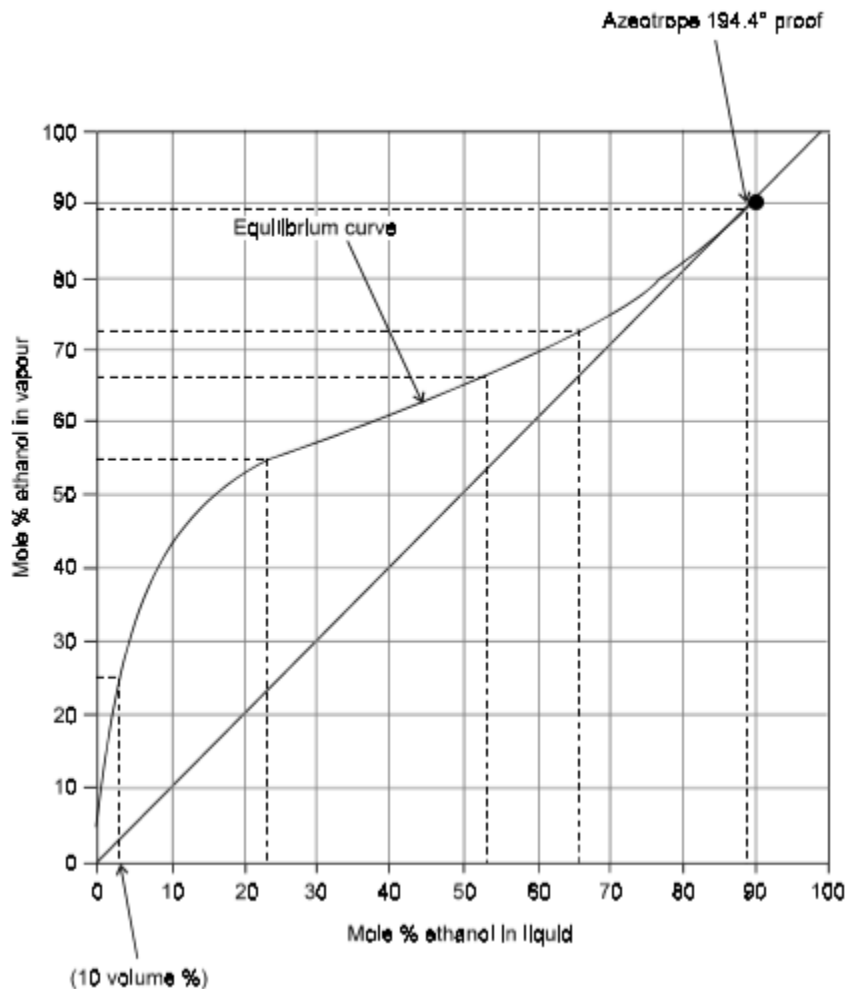


Figure 10. Vapor/liquid equilibrium for the ethanol/water system at atmospheric pressure. [8]

In both Figures 8 and 9, the recovery (X_d) of ethanol is approaching this pinch point indicating the need for an infinitely large column. Recoveries obtained in this experiment are highly unlikely if not impossible. The high efficiency and high recovery can be accounted for by experimental error and is discussed in Appendix II.

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Appendix I: Data Tabulation and Graphs*Table 4. Calibration curve raw data where RI means “Refractive Index”*

ml of Ethanol	ml DI Water	g Ethanol	g DI water	wt %	mol ethanol	mol water	mol fraction Ethanol	RI Trial 1	RI Trial 2	RI Trial 3	Average RI
0	10	0	9.97	0.0000	0.0000	0.5533	0.0000	0	0	0	0.0000
1	9	0.789	8.973	8.0824	0.0171	0.4979	0.0333	1.3357	1.3344	1.3374	1.3358
2	8	1.578	7.976	16.5166	0.0343	0.4426	0.0718	1.34	1.3393	1.3406	1.3400
3	7	2.367	6.979	25.3263	0.0514	0.3873	0.1171	1.3489	1.3493	1.3495	1.3492
4	6	3.156	5.982	34.5371	0.0685	0.3320	0.1711	1.3541	1.354	1.3545	1.3542
5	5	3.945	4.985	44.1769	0.0856	0.2766	0.2364	1.3588	1.3597	1.3591	1.3592
6	4	4.734	3.988	54.2765	0.1028	0.2213	0.3171	1.3641	1.3644	1.3644	1.3643
7	3	5.523	2.991	64.8696	0.1199	0.1660	0.4194	1.366	1.3662	1.3661	1.3661
8	2	6.312	1.994	75.9933	0.1370	0.1107	0.5532	1.3636	1.3632	1.3636	1.3635
9	1	7.101	0.997	87.6883	0.1541	0.0553	0.7359	1.3631	1.3633	1.3633	1.3632
10	0	7.89	0	100.0000	0.1713	0.0000	1.0000	1.3624	1.3622	1.3623	1.3623

Table 5: Raw data collected for the boil-up rate calculations

Power(kW)	volume (ml)			time (s)		
	top S1	top S2	V _{ave} (ml)	top S1	top S2	T _{ave} (s)
0.5	12.7	10.5	11.6	46	35.74	40.87
0.75	11	11.2	11.1	15.62	17.91	16.765
1	13	13.3	13.15	14.22	13.5	13.86
1.25	13	13.7	13.35	10.35	12.4	11.375

Table 6: Calculated boil-up rate from the data in table 2.

Power(kW)	Boilup Rate (L/hr)	Overall dP	log(dP)	log(boilup rate)
0.5	1.021776364	67	1.826074803	0.009355852
0.75	2.383537131	68	1.832508913	0.377221922
1	3.415584416	74	1.86923172	0.533465023
1.25	4.225054945	105	2.021189299	0.625832361

Table 7: Literature VLE data for water/ethanol [5]

x(EtOH)	y(EtOH)	x(water)	y(water)	T (°C)
0	0	1	1	100
0.0190	0.1700	0.9810	0.8300	95.5
0.0721	0.3891	0.9279	0.6109	89
0.0966	0.4375	0.9034	0.5625	86.7
0.1238	0.4704	0.8762	0.5296	85.3
0.1661	0.5089	0.8339	0.4911	84.1
0.2337	0.5445	0.7663	0.4555	82.7
0.2608	0.5580	0.7392	0.4420	82.3
0.3273	0.5826	0.6727	0.4174	81.5
0.3965	0.6122	0.6035	0.3878	80.7
0.5198	0.6599	0.4802	0.3401	79.7
0.5732	0.6841	0.4268	0.3159	79.3
0.6763	0.7385	0.3237	0.2615	78.74
0.7472	0.7815	0.2528	0.2185	78.41
0.8943	0.8943	0.1057	0.1057	78.15
1	1	0	0	78.3

Table 8. McCabe Thiele data for reflux ratio 5:1 Part B

Literature VLE Data		Equilibrium $y = x$		Rectifying	
x	y	x	y	x	y
0	0	0	0	0	0.145167
0.019	0.17	0.019	0.019	0.019	0.161
0.0721	0.3891	0.0721	0.0721	0.0721	0.20525
0.0966	0.4375	0.0966	0.0966	0.0966	0.225667
0.1238	0.4704	0.1238	0.1238	0.1238	0.248334
0.1661	0.5089	0.1661	0.1661	0.1661	0.283584
0.2337	0.5445	0.2337	0.2337	0.2337	0.339917
0.2608	0.558	0.2608	0.2608	0.2608	0.3625
0.3273	0.5826	0.3273	0.3273	0.3273	0.417917
0.3965	0.6122	0.3965	0.3965	0.3965	0.475584
0.5198	0.6599	0.5198	0.5198	0.5198	0.578334
0.5732	0.6841	0.5732	0.5732	0.5732	0.622834
0.6763	0.7385	0.6763	0.6763	0.6763	0.70875
0.7472	0.7815	0.7472	0.7472	0.7472	0.767834
0.8943	0.8943	0.8943	0.8943	0.8943	0.890417
1	1	1	1	1	0.9785

Table 9. McCabe Thiele data for reflux ratio 10:1 Part B

Literature VLE Data		Equilibrium $y = x$		Rectifying	
x	y	x	y	x	y
0.0000	0.0000	0.0000	0.0000	0.0000	0.0799
0.0190	0.1700	0.0190	0.0190	0.0190	0.0971
0.0721	0.3891	0.0721	0.0721	0.0721	0.1454
0.0966	0.4375	0.0966	0.0966	0.0966	0.1677
0.1238	0.4704	0.1238	0.1238	0.1238	0.1924
0.1661	0.5089	0.1661	0.1661	0.1661	0.2309
0.2337	0.5445	0.2337	0.2337	0.2337	0.2923
0.2608	0.5580	0.2608	0.2608	0.2608	0.3169
0.3273	0.5826	0.3273	0.3273	0.3273	0.3774
0.3965	0.6122	0.3965	0.3965	0.3965	0.4403
0.5198	0.6599	0.5198	0.5198	0.5198	0.5524
0.5732	0.6841	0.5732	0.5732	0.5732	0.6009
0.6763	0.7385	0.6763	0.6763	0.6763	0.6947
0.7472	0.7815	0.7472	0.7472	0.7472	0.7591
0.8943	0.8943	0.8943	0.8943	0.8943	0.8929
1.0000	1.0000	1.0000	1.0000	1.0000	0.9889

Table 10. Raw data for boil-up rate calculation in Part B

	Reflux ratio				T (°C)	
	10 to 1	5 to 1	10 to 1	5 to 1	10 to 1	
Sample	Volume (ml)		Time (s)		process	column
S1 top	7.8	8	11.15	15.31	92.7	82.4
S2 top	8	8.2	12	13.87	5 to 1	
S3 top	8.8	7.2	13.25	11.91	93.1	82.4

Table 11. Refractive index raw data for Part B reflux ratio 10/1 with calculated mol fraction

REFLUX RATIO 10/1		Trial 1 RI	Trial 2 RI	Trial 3 RI	Average RI	mol fraction ethanol
	sample1	1.3649	1.365	1.365	1.3650	0.8821
	sample 2	1.3648	1.365	1.365	1.3649	0.8835
	sample 3	1.3652	1.3654	1.3652	1.3653	0.8696
	bottoms	1.3377	1.3376	1.3379	1.3377	0.0177

Table 12. Refractive index raw data for Part B reflux ratio 5/1 with calculated mol fraction

REFLUX RATIO 5/1		Trial 1 RI	Trial 2 RI	Trial 3 RI	Average RI	mol fraction ethanol
	sample 1	1.3651	1.3648	1.3651	1.3650	0.8808
	sample 2	1.3654	1.3656	1.3656	1.3655	0.8585
	sample 3	1.3651	1.3654	1.365	1.3652	0.8738
	bottoms	1.3382	1.338	1.3382	1.3381	0.0122

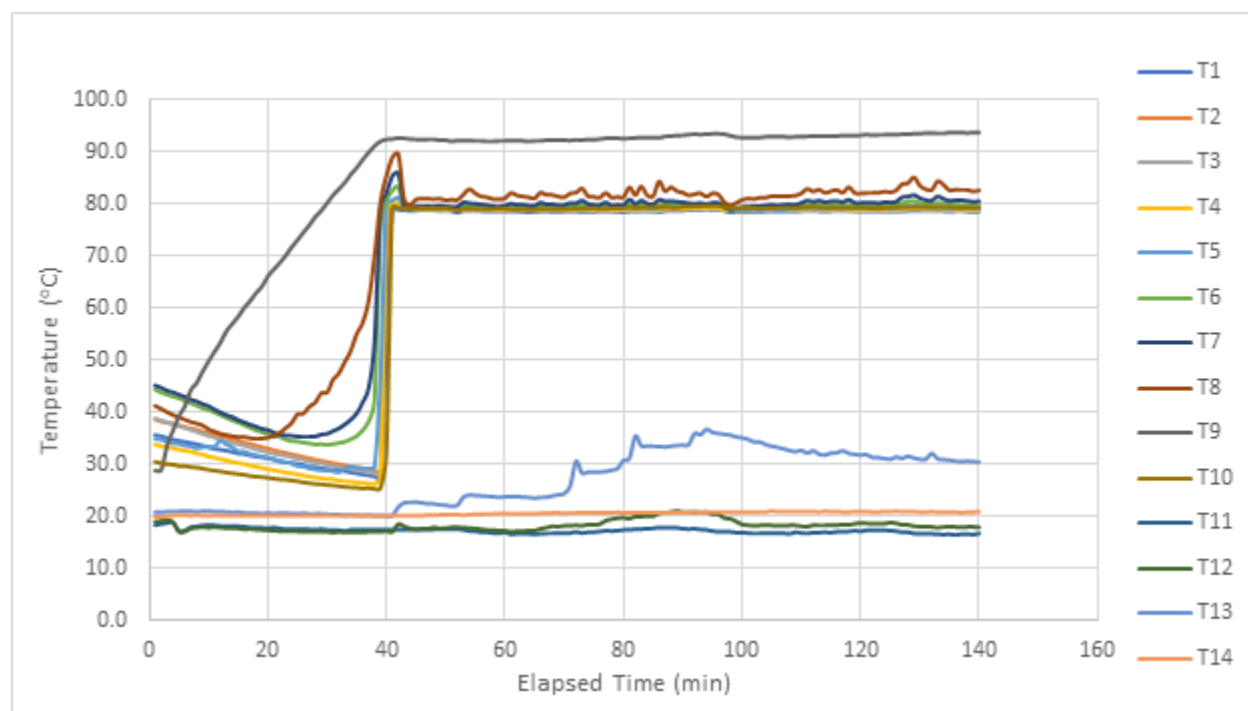


Figure 11: Detailed presentation of the temperatures of the column vs the elapsed time.

Appendix II: Error Analysis

An accurate calibration curve is essential to the success of this experiment. The solutions to construct the calibration curve are prepared by measuring volumes of each component and converting these volumes into molar concentrations. These conversions may not be as accurate as actually weighing out the amount of each component for each solution. This may account for the higher variance in the best fit curve for the molar concentration of ethanol curve vs the weight percent ethanol curve. Since the molar concentration curve is primarily used to convert the refractive index of each sample to molar fraction of ethanol, a higher level of error is present. It can be surmised that the great level of error encountered in this experiment can be attributed to a calibration curve that is less than ideal.

Another source of error could be human error in the correct handling of the sample collection: recording of the time needed for the sampling of the overhead, as well as the removal of the impurities before the flow rates are recorded.

The refractive index of each calibration solution is measured three times, and the average is calculated along with the standard deviation and relative standard deviation. See Table 13. The small relative standard deviations indicate that the refractive indices measured are precise, though not necessarily accurate.

Table 13. Calibration curve refractive index error analysis with standard deviation and relative standard deviation. RI means "Refractive Index".

mol fraction Ethanol	RI Trial 1	RI Trial 2	RI Trial 3	Average RI	Standard Deviation	Relative Standard Deviation
0.0000	0	0	0	0.0000	0.0000	
0.0333	1.3357	1.3344	1.3374	1.3358	0.0015	0.1126
0.0718	1.34	1.3393	1.3406	1.3400	0.0007	0.0486
0.1171	1.3489	1.3493	1.3495	1.3492	0.0003	0.0226
0.1711	1.3541	1.354	1.3545	1.3542	0.0003	0.0195
0.2364	1.3588	1.3597	1.3591	1.3592	0.0005	0.0337
0.3171	1.3641	1.3644	1.3644	1.3643	0.0002	0.0127
0.4194	1.366	1.3662	1.3661	1.3661	0.0001	0.0073
0.5532	1.3636	1.3632	1.3636	1.3635	0.0002	0.0169
0.7359	1.3631	1.3633	1.3633	1.3632	0.0001	0.0085
1.0000	1.3624	1.3622	1.3623	1.3623	0.0001	0.0073

Other sources of error can come from human error in measuring or in contaminating samples, or equipment error including the column, reboiler heat source, or refractometer inaccuracies or malfunctions, though none was noted throughout the experiment.

Appendix III: Sample Calculations**Calibration Curve****Weight percent calculations:**

$$\text{Volume} \times \text{density} = \text{mass}$$

$$4 \text{ ml ethanol} * \frac{0.789 \text{ g ethanol}}{1 \text{ ml ethanol}} = 3.156 \text{ g ethanol}$$

$$6 \text{ ml water} * \frac{.997 \text{ g water}}{1 \text{ ml water}} = 5.982 \text{ g water}$$

$$\frac{\text{mass of ethanol}}{\text{total mass}} * 100 = \text{wt. \% ethanol}$$

$$\frac{3.156 \text{ g ethanol}}{3.156 \text{ g ethanol} + 5.982 \text{ g water}} * 100 = 34.54 \text{ wt. \% ethanol}$$

Mol fraction calculations:

$$\frac{\text{mass}}{\text{molecular weight}} = \text{mol}$$

$$\frac{3.156 \text{ g ethanol}}{46.07 \frac{\text{g ethanol}}{\text{mol ethanol}}} = .0685 \text{ mol ethanol}$$

$$\frac{5.981 \text{ g water}}{18.02 \frac{\text{g water}}{\text{mol water}}} = .3319 \text{ mol water}$$

$$\frac{\text{mol ethanol}}{\text{mol ethanol} + \text{mol water}} = \text{mol fraction of ethanol}$$

$$\frac{.0685 \text{ mol ethanol}}{.0685 \text{ mol ethanol} + .3319 \text{ mol water}} = .1711$$

Part A: Effect of Boil-Up Rate and Overall Column Efficiency

Boil-up rate sample calculation:

$$\text{boil-up rate} = \frac{V_{ave}}{T_{ave}} = \frac{11.6 * 3600}{40.87 * 1000} = 1.0218 \text{ l/hr}$$

The calibration curve is used to find the molar compositions of the overhead and bottoms as presented in the sample calculations above.

Calculations for the number of theoretical trays N at 0.5 kW power:

In the distillate:

$$x(\text{EtOH})_d = 0.9177$$

$$x(\text{water})_d = 1 - 0.9177 = 0.0883$$

In the bottoms:

$$x(\text{EtOH})_b = 0.0165$$

$$x(\text{water})_b = 1 - 0.0165 = 0.9835$$

To find the vapor fraction of ethanol the literature VLE data (table 7 Appendix I) is used to interpolate with the $x(\text{EtOH})$ from the overhead:

$$y_{\text{EtOH}} = y_1 + (x_{\text{EtOH}} - x_1) \frac{y_2 - y_1}{x_2 - x_1}$$

$$y_{\text{EtOH}} = 0.8943 + (0.9117 - 0.8943) * \frac{1 - 0.8943}{1 - 0.8943} = 0.9177$$

$$\alpha_{\text{EtOH}} = \frac{\frac{y_{\text{EtOH}}}{x_{\text{EtOH}}}}{(1 - y_{\text{EtOH}})/(1 - x_{\text{EtOH}})}$$

$$\alpha = \frac{0.9117}{0.9117} / \left(\frac{1 - 0.9117}{1 - 0.9117} \right) = 1$$

Number of theoretical trays N are computed:

$$N = \frac{\log \left[\left(\frac{x_{\text{EtOH}}}{x_{\text{H}_2\text{O}}} \right)_d \left(\frac{x_{\text{H}_2\text{O}}}{x_{\text{EtOH}}} \right)_b \right]}{\log(\alpha_{\text{EtOH}/\text{H}_2\text{O}})_{\text{avg}}} - 1$$

$$\alpha_{\text{ave}} = \text{sqrt}(1 * 0.0969) = 0.3159$$

$$N = \text{abs} \left\{ \frac{\log \left(\frac{0.9117}{0.0883} * \frac{0.9835}{0.0165} \right)}{\log(0.3159)} - 1 \right\} = 6.57$$

$$\text{Efficiency} = \frac{6.57}{8} * 100 = 82.17\%$$

Part B: Distillation at a Constant Reflux Ratio**Using calibration curve to calculate mol fraction of ethanol:**

$$-0.0766 * x_{ethanol} + 0.0992 * x_{ethanol} + 1.3366 = y$$

Where 'y' is the refractive index of the sample.

Plugging in 'y'-value and subtracting to the other side yields:

$$-0.0766 * x_{ethanol} + 0.0992 * x_{ethanol} + 1.3366 - y = 0$$

In which GoalSeek in Excel can be used to solve the equation.

For Reflux Ratio 10/1:

Refractive index for sample 1 from the distillate is 1.3560.

$$-0.0766 * x_d + 0.0992 * x_d + 1.3366 - 1.3560 = 0$$

Using GoalSeek, the larger of the two solutions is chosen because the product on the top of the column is richer in ethanol than the bottom. The smaller of the two solutions would be chosen for the bottoms.

$$X_d = .8821$$

This is repeated for all top samples and then the average is taken to find the average molar fraction present in the distillate at that reflux ratio. This is again repeated for the bottoms sample.

Boil-up rate Calculations:

$$\text{Boil-Up Rate} = \frac{\text{Volume Collected}}{\text{Time}}$$

For 10/1 reflux ratio, sample 1:

$$\frac{7.8 \text{ ml} * 3600 \frac{\text{sec}}{\text{hour}}}{11.15 \text{ sec} * \frac{1000 \text{ ml}}{1 \text{ l}}} = 2.5184 \frac{\text{l}}{\text{hour}}$$

The average of all boil-up rates for each reflux ratio is used for McCabe Thiele analysis.

Rectifying Line:

$$y_{n+1} = \left(\frac{L}{V}\right) x_n + \left(1 - \frac{L}{V}\right) x_d$$

$$\frac{L}{V} = \frac{L}{L + D} = \frac{\frac{L}{D}}{\frac{L}{D} + 1}$$

$$\frac{L}{D} = \text{Reflux Ratio}$$

$$y_{n+1} = \left(\frac{\frac{L}{D}}{\frac{L}{D} + 1} \right) x_n + \left(1 - \frac{\frac{L}{D}}{\frac{L}{D} + 1} \right) x_d$$

For Reflux Ratio 10/1:

$$y_{n+1} = \left(\frac{10}{10 + 1} \right) x_n + \left(1 - \frac{10}{10 + 1} \right) 0.8784$$

$$y_{n+1} = \left(\frac{10}{11} \right) x_n + 0.07986$$

For Reflux Ratio 5/1:

$$y_{n+1} = \left(\frac{5}{5 + 1} \right) x_n + \left(1 - \frac{5}{5 + 1} \right) 0.871$$

$$y_{n+1} = \left(\frac{5}{6} \right) x_n + 0.1452$$

Appendix IV: Individual Team Contributions

NAME: Kellen Krupa

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Present on both days
Pre-Lab Editing	0.5	
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)	1	
Final Lab Editing	2	Putting sections together and reviewing edit
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)	7	Research and writing
Results (final lab)		
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)	0.5	Formatted citations and sources
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	2	
Total	20	

NAME: Cheryl Slykas

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Present both lab periods
Pre-Lab Editing	2	
Experimental Objective (Prelab)	.5	
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)	1.5	
Project Safety Evaluation (prelab)		
Final Lab Editing	.5	

Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	3	
Discussion (final lab)	3	
Conclusion (final lab)		
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)	12	So many hours
Error Analysis (final lab)	1	
Sample Calculations (final Lab)	2	
Power Point Presentation	.5	
Total	34	

NAME: Adisa Hadzic

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Accomplished pre-lab and lab experiment.
Pre-Lab Editing	1	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	2	Labeled all parts of the apparatus
Materials & Supplies (prelab)	1	Went over picture and labeled with their functions
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1	Final corrections to lab
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	12	Results and making sure calculations are correct (Exp A)
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	5	Experiment A
Error Analysis (final lab)	0.1	
Sample Calculations (final Lab)	2	
Power Point Presentation	1	
Total	32.1	

NAME: Marium Murtaza

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	7	Accomplished pre-lab and lab experiment.
Pre-Lab Editing	1	editing
Experimental Objective (Prelab)	3	Objective of part a and part b of lab
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1	Final corrections to lab
Executive Summary (final lab)	3	Summary Part
Introduction (Final lab)	3	Intro and uses in real life
Theory (Final Lab)	0	
Results (final lab)		
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	0	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	2	Safety and Materials part
Total	20	